ZACCHAEUS ADEDEJI, GMNSE

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PROFESSIONAL STATEMENT

GMNSE-registered Mechanical Engineer specializing in HVAC design, VRF/VRV systems, and BIM coordination for commercial and mixed-use buildings. Experienced across the full project lifecycle, from load calculation and Revit modelling through site supervision and commissioning, cutting modelling time, rework, and system faults through reusable design components and close team and contractor coordination.

SKILLS

- Revit & SolidWorks
- AI/LLM Prompt Engineering
- Client & Stakeholder Communication
- Energy Optimization
- Mechanical Diagnostics
- Time Management
- Team Support
- HVAC System Design (DX, VRV/VRF, HAP)
- Site Supervision & Commissioning
- Machine Design
- Problem Solving
- Analytical Thinking
- Attention to Detail
- Organisation

PROFESSIONAL EXPERIENCE

HVAC Designer *January 2026 – Present*

View Scope Engineering Services Limited, Lagos, Nigeria

- Designed an energy-efficient HVAC system for a commercial office building with 800+ indoor units, optimizing energy use and indoor comfort.
- Supervised installation and commissioning of 36 DX indoor units (highwall, cassette, and ceiling-concealed) across a 2-floor retrofit, completed in 5 weeks via close contractor coordination and on-site issue resolution.
- Developed reusable Revit components, improving modelling efficiency by 40%.
- Coordinated mechanical layouts with multidisciplinary teams, reducing on-site rework by 30%.
- Prepared technical documentation and delivered project proposals to support informed stakeholder decisions.
- Designed and modeled DX and VRV systems for a range of cooling spaces — hotel, residential, and mixed-use — selecting equipment using HAP and Revit.

KEY PROJECTS

AI-Driven HVAC & MEP Design Automation *2024 – 2025*

Final Year Project, Obafemi Awolowo University — Team of 2

- Co-designed the system architecture and end-to-end workflow for an AI-assisted tool automating HVAC/MEP design for 1–2 bedroom residential buildings in Nigeria — from drawing input through equipment selection to Bill of Quantities generation.
- Led prompt engineering for the tool's LLM layer, directing AI-driven equipment recommendations (compressors, AHUs) from building load and environmental inputs.
- Contributed to a drawing-parsing pipeline converting architectural PDFs/images into CLTD/CLF-based load inputs, cutting manual interpretation time by an estimated 60–70%.
- Benchmarked the AI-assisted workflow against manual design methods and shaped the tool's outputs around Nigerian building practice, showing gains in speed and consistency for local consulting and independent engineers.

Tools & Methods: *Prompt Engineering, System/Workflow Architecture, CLTD/CLF Load Calculations, BOQ Automation*

INTERNSHIP EXPERIENCE

IT Student *April 2024 – September 2024*

Division of Works and Maintenance Service, Ile-Ife, Nigeria

- Supported HVAC system installation, servicing, and troubleshooting for 5+ commercial and residential systems, improving reliability and efficiency.
- Assisted with mechanical maintenance tasks, including engine, clutch, and brake servicing, reducing recurring faults by 60%.
- Applied cooling load and airflow analysis to optimize system performance.

IT Student *June 2023 – August 2023*

Central Technological and Laboratory Workshops, Ile-Ife, Nigeria

- Assisted in belt design and machine installation, improving operational efficiency and reducing slippage by 25%.
- Contributed to fabrication of a plantain slicing machine, completing assembly 40% faster and increasing processing capacity.
- Operated workshop machinery safely, supporting machining, welding, and component fabrication.

EDUCATION & CERTIFICATIONS

Obafemi Awolowo University, Ile-Ife, Nigeria | B.Sc. Mechanical Engineering — 2025

- Graduate Member, Nigerian Society of Engineers (GMNSE)